

Lesson 3: Discrete Probability

Main reference : Discrete Mathematics and Its Applications, 7th edition, Kenneth H. Rosen
Additional references: Probability and Statistics for Computer Scientists, Second Edition, Michael Baron
and Introduction to Probability, Second Edition, Charles Grinstead and J. Laurie Snell

In computer science, probability theory plays an important role in the study of the complexity of algorithms. Specifically, the average-case complexity of algorithms can be determined using ideas and techniques from probability theory. Probability theory can also help in answering questions that involve uncertainty, such as determining whether we should reject an incoming mail message as spam based on the words that appear in the message. We now define some basic concepts in discrete probability.

1 Finitely Many, Equally Likely

Definition 1 An *experiment* is a procedure that yields one of a given set of possible outcomes. The *sample space* of the experiment is the set of possible outcomes. An *event* is a subset of the sample space.

Definition 2 (Laplace's Definition of Probability)

If S is a finite nonempty sample space of equally likely outcomes, and E is an event, which is a subset of S , then the **probability** of E is

$$p(E) = \frac{|E|}{|S|}$$

Since E is a subset of S , then it follows that $0 \leq p(E) \leq 1$.

Example 1

1. An urn contains four blue balls and five red balls. What is the probability that a ball chosen from the urn is blue?

Solution: There are nine equally likely possible outcomes, and four of these possible outcomes produce a blue ball. Hence, $\frac{4}{9}$.

2. What is the probability that when two distinguishable dice are rolled, the sum of the numbers on the two dice is 7?

Solution: There are 36 equally likely possible outcomes. Six of these sum to 7. Hence, $\frac{6}{36} = \frac{1}{6}$.

3. In a lottery, a player wins a large prize when they pick four digits that match in correct order the four digits selected by a random mechanical process.

- (a) What is the probability that a player wins the prize?

Solution: By the product rule, there are $10^4 = 10000$ ways to pick the prize. Since there is only one way to pick the correct digits, the probability of winning is 0.0001.

- (b) If a smaller prize can be won by matching three out of four digits, what is the probability that a player wins the smaller prize?

Solution: If exactly three digits are matched, one of four digits must be incorrect and the other three digits must be correct. The incorrect digit can be any of the 4. For that incorrect digit, there are 9 possible choices. hence, there are a total of $4(9)=36$ possible ways to choose four digits that match exactly three of the winning four digits. The probability of winning the small prize is $36/10000 = 0.0036$.

4. In a lottery, the grand prize is awarded to people who correctly choose a set of six numbers out of $\{1, 2, 3, \dots, 39, 40\}$. What is the probability that a person wins the prize?

Solution: The total number of ways to choose six numbers out of 40 is $C(40, 6) = \frac{40!}{34!6!} = 3,838,380$. There is only one winning combination. Hence, the probability of winning is $1/3,838,380 \approx 0.00000026$.

The Probability of Complements and Unions of Events

We can use counting techniques to find the probability of events derived from other events.

Theorem 1 Let E be an event in sample space S . The probability of the event $\bar{E} = S - E$, the complementary event of E , is given by

$$p(\bar{E}) = 1 - p(E)$$

Example 2 A sequence of 10 bits is chosen randomly. What is the probability that at least one of these bits is 0?

Solution: There are 2^{10} ways to pick the 10-bit sequence. Let E be the event when all of the bits are 1s. Then $\bar{E}$ is the event that at least one of the 10 bits is 0. There is only one 10-bit sequence that are all ones. Hence,

$$p(\bar{E}) = 1 - p(E) = 1 - \frac{1}{2^{10}} = \frac{1023}{1024}$$

Theorem 2 Let E_1 and E_2 be events in the sample space S . Then,

$$p(E_1 \cup E_2) = p(E_1) + p(E_2) - p(E_1 \cap E_2)$$

Example 3 What is the probability that a positive integer selected from 1 to 100 is divisible by either 2 or 5?

Solution: Let E_1 be the event that the integer is divisible by 2 and E_2 be the event that the integer is divisible by 5. Then $E_1 \cup E_2$ is the event that the integer is divisible by either 2 or 5, and $E_1 \cap E_2$ is the event that the integer is divisible by both 2 and 5 (that is, divisible by 10). Since $|E_1| = 50$, $|E_2| = 20$ and $|E_1 \cap E_2| = 10$, then

$$\begin{aligned} p(E_1 \cup E_2) &= p(E_1) + p(E_2) - p(E_1 \cap E_2) \\ &= \frac{50}{100} + \frac{20}{100} - \frac{10}{100} \\ &= \frac{60}{100} = \frac{3}{5} \end{aligned}$$

2 Outcomes, Not Equally Likely

Many experiments have outcomes that are not equally likely. For instance, a coin may be biased so that it comes up heads more often than tails. How can we model the likelihood of events in such situations? In the following definitions, we show how to define probabilities of outcomes of experiments where these outcomes may not be equally likely.

Definition 3 Let S be the sample space of an experiment with a finite number of outcomes. We assign a probability to each outcome s , denoted as $p(s)$ so that,

(i) $0 \leq p(s) \leq 1$ for each $s \in S$, and

(ii) $\sum_{s \in S} p(s) = 1$

The function $p : S \rightarrow [0, 1]$ is called a **probability distribution**.

Example 4 What are the probabilities for the outcomes heads (H) or tails (T) on a fair coin? What are the probabilities for the outcomes of heads or tails when the coin is biased that heads comes up twice as often as tails?

Solution: Note that in both cases, $p(H) + p(T) = 1$. For the fair coin, $p(H) = p(T)$. Thus, $p(H) = p(T) = \frac{1}{2}$. For the biased coin, $p(H) = 2p(T)$. So $p(H) = \frac{2}{3}$ and $p(T) = \frac{1}{3}$.

Note that we can still use Definition 3 to assign probabilities for outcomes of an experiment which are equally likely.

Definition 4 Let S be a sample space of an experiment with n possible outcomes which are equally likely. The **uniform distribution** assigns the probability $\frac{1}{n}$ to each element of S .

We now define the probability of an event as the sum of the probabilities of the outcomes in the event.

Definition 5 The **probability of the event** E is the sum of probabilities of the outcomes in E , given by

$$p(E) = \sum_{s \in E} p(s)$$

Example 5 Suppose that a dice is biased so that 3 appears twice as often as each other number, but the other five outcomes are equally likely. What is the probability that an odd number appears when we roll this dice?

Solution: Let E be the event of rolling an odd number, that is, $E = \{1, 3, 5\}$. Note that $p(1) + p(2) + p(3) + p(4) + p(5) + p(6) = 1$. Also, $p(1) = p(2) = p(4) = p(5) = p(6)$ while $p(3) = 2p(1)$. Thus, the individual probabilities are $p(3) = \frac{2}{7}$, while $p(1) = p(2) = p(4) = p(5) = p(6) = \frac{1}{7}$. Hence,

$$\begin{aligned} p(E) &= p(1) + p(3) + p(5) \\ &= \frac{1}{7} + \frac{2}{7} + \frac{1}{7} \\ &= \frac{4}{7} \end{aligned}$$

Theorem 2 still holds for events with varying probabilities. But if the events E_1 and E_2 are disjoint, then $p(E_1 \cap E_2) = 0$, implying that $p(E_1 \cup E_2) = p(E_1) + p(E_2)$. We can extend this formula for the probability of the union of pairwise disjoint events, where pairwise disjoint means that every pair of events in the collection are disjoint.

Theorem 3 If $E_1, E_2, \dots$ is a sequence of pairwise disjoint events in a sample space S , then

$$p\left(\bigcup_i E_i\right) = \sum_i p(E_i).$$

3 Conditional Probabilities

Suppose that a fair coin is flipped three times. Moreover, suppose we know that the first flip comes up tails, call this event F . What is the probability of the event E , that an odd number of tails appears? This probability is called the conditional probability of E given F . To find the conditional probability of E given F , we use F as the sample space. For an outcome from E to occur, this outcome must also belong to $E \cap F$. This is the motivation for this definition:

Definition 6 Let E and F be events with $p(F) > 0$. The **conditional probability** of E given F , denoted as $p(E | F)$ is defined as:

$$p(E | F) = \frac{p(E \cap F)}{p(F)}$$

Example 6 Imagine a couple with two children, each of whom is equally likely to be a boy or a girl. Now suppose you are given the information that one is a boy. What is the probability that the other child is a boy?

Solution: There are four possibilities, BB , BG , GB and GG , which are equally likely to happen. Since there is information that one is a boy, the possible outcomes are BB , BG and GB . Let E be the event that the two children are both boys, and let F be the event that one of the two children is a boy. Then the probability of E given F is:

$$p(E | F) = \frac{p(E \cap F)}{p(F)} = \frac{\frac{1}{4}}{\frac{3}{4}}$$

Example 7 An urn contains 5 blue balls and 7 gray balls. Two balls are chosen at random, one after the other, without replacement.

1. Compute for the following probabilities:

- (a) the probability that both balls are blue,
- (b) the probability that the first ball is blue but the second is not blue,
- (c) the probability that the first ball is not blue and the second ball is blue, and
- (d) the probability that neither ball is blue.

Solution: Let S denote the sample space of all possible choices of two balls from the urn, let B_1 be the event that the first ball is blue, and B_2 be the event that the second ball is blue. Then, $\overline{B_1}$ is the event that the first ball is not blue and $\overline{B_2}$ is the event that the second ball is not blue. Need to compute $p(B_1 \cap B_2)$, $p(B_1 \cap \overline{B_2})$, $p(\overline{B_1} \cap B_2)$, and $p(\overline{B_1} \cap \overline{B_2})$.

Since $|S| = 12$ (5 of which are blue and 7 are gray), then

$$p(B_1) = \frac{5}{12}, \quad p(\overline{B_1}) = \frac{7}{12}$$

If the first is blue, then the urn would contain 4 blue balls and 7 gray balls. So,

$$p(B_2 | B_1) = \frac{4}{11}, \quad p(\overline{B_2} | B_1) = \frac{7}{11}$$

It follows that

$$p(B_1 \cap B_2) = p(B_2 \cap B_1) = p(B_2 | B_1)p(B_1) = \frac{4}{11} \left(\frac{5}{12} \right) = \frac{20}{132}$$

and

$$p(B_1 \cap \overline{B_2}) = p(\overline{B_2} \cap B_1) = p(\overline{B_2} | B_1)p(B_1) = \frac{7}{11} \left(\frac{5}{12} \right) = \frac{35}{132}.$$

Now, if the first ball is not blue, then the urn would have 5 blue balls and 6 gray balls. So,

$$p(B_2 | \overline{B_1}) = \frac{5}{11}, \quad p(\overline{B_2} | \overline{B_1}) = \frac{6}{11}.$$

It follows that,

$$p(\overline{B_1} \cap B_2) = p(B_2 \cap \overline{B_1}) = p(B_2 | \overline{B_1})p(\overline{B_1}) = \frac{5}{11} \left(\frac{7}{12} \right) = \frac{35}{132},$$

and

$$p(\overline{B_1} \cap \overline{B_2}) = p(\overline{B_2} \cap \overline{B_1}) = p(\overline{B_2} | \overline{B_1})p(\overline{B_1}) = \frac{6}{11} \left(\frac{7}{12} \right) = \frac{42}{132}.$$

2. What is the probability that the second ball is blue?

Solution: The event that the second ball is blue can occur in one of two mutually exclusive ways: either the first ball is blue or is gray, then followed by the second ball that is blue. Hence,

$$\begin{aligned} p(B_2) &= p(B_2 \cap B_1) + p(B_2 \cap \overline{B_1}) \\ &= \frac{20}{132} + \frac{35}{132} \\ &= \frac{55}{132} = \frac{5}{12}. \end{aligned}$$

3. What is the probability that at least one of the balls is blue?

Solution: The event that neither ball is blue is the complement of the event that at least one of the balls is blue, so

$$\begin{aligned} 1 - p(\overline{B_1} \cap \overline{B_2}) &= 1 - \frac{42}{132} \\ &= \frac{90}{132}. \end{aligned}$$

Also,

$$\begin{aligned} p(B_1 \cap B_2) + p(B_1 \cap \overline{B_2}) + p(\overline{B_1} \cap B_2) &= \frac{20}{132} + \frac{35}{132} + \frac{35}{132} \\ &= \frac{90}{132}. \end{aligned}$$

Independent Sets

Consider a fair coin. Let E be the event that a head is obtained on the first toss and let F be the event that a head is obtained on the second toss. If the coin is tossed randomly both times, events E and F should be independent events, in the sense that $p(E | F) = p(E)$ and $p(F | E) = p(F)$. Since $p(E | F)p(F) = p(E \cap F) = p(F | E)p(E)$, this leads to the following definition for independent events:

Definition 7 *The events E and F are **independent** if and only if*

$$p(E \cap F) = p(E) \cdot p(F)$$

Example 8 *A coin is biased so that the probability of heads is 0.6. Suppose the coin is tossed twice. Additionally, one toss will not affect the results of any succeeding coin tosses. Thus it can be assumed that coin tosses are independent of each other.*

1. What is the probability of obtaining two heads?
2. What is the probability of obtaining one head?
3. What is the probability of obtaining no heads?
4. What is the probability of obtaining at least one head?

Solution: Let S be the sample space of the two coin tosses, consisting of $\{HH, HT, TH, TT\}$, each of which are not equally likely. Let E be the event that a head is obtained on the first toss and F be the event that a head is obtained on the second toss. Since both events are assumed independent, $p(E) = p(F) = 0.6$.

The probability of obtaining two heads is $p(E \cap F)$. By definition of independent sets,

$$\begin{aligned} p(E \cap F) &= p(E) \cdot p(F) \\ &= (0.6)(0.6) \\ &= 0.36 \end{aligned}$$

Next, one head can be obtained in two mutually exclusive ways: head on the first toss and tail on the second toss; tail on the first toss and head on the second toss. Hence, the probability of obtaining exactly one head is:

$$\begin{aligned} p(\text{one head}) &= p((E \cap \bar{F}) \cup (\bar{E} \cap F)) \\ &= p(E)p(\bar{F}) + p(\bar{E})p(F) \\ &= (0.6)(0.4) + (0.4)(0.6) \\ &= 0.48 \end{aligned}$$

The probability of obtaining no heads is $p(\bar{E} \cap \bar{F})$, computed as

$$\begin{aligned} p(\bar{E} \cap \bar{F}) &= p(\bar{E})p(\bar{F}) \\ &= (0.4)(0.4) \\ &= 0.16 \end{aligned}$$

Finally, by complement rule, the probability of obtaining at least one head is to subtract the probability of obtaining no heads from 1. That is,

$$\begin{aligned} p(\text{at least one head}) &= 1 - p(\text{no heads}) \\ &= 1 - p(\bar{E} \cap \bar{F}) \\ &= 1 - 0.16 \\ &= 0.84 \end{aligned}$$

Example 9 Consider the events in Example 6. Are the events E , that the two children are boys, and F , that one of the two children is a boy, independent?

Solution: Recall that $p(E) = \frac{1}{4}$, $p(F) = \frac{3}{4}$ and $p(E \cap F) = \frac{1}{4}$. Hence, $p(E \cap F) \neq p(E)p(F)$, so the events E and F are not independent.

We can also define the independence of a collection of events. However, we shall define two types of independence, given in the following:

Definition 8 The events $E_1, E_2, \dots, E_n$ are **pairwise independent** if and only if $p(E_i \cap E_j) = p(E_i)p(E_j)$ for all pairs of integers i and j with $1 \leq i < j \leq n$. These events are **mutually independent** if and only if $p(E_1 \cap E_2 \cap \dots \cap E_n) = p(E_1)p(E_2) \dots p(E_n)$.

4 Bayes' Theorem

Oftentimes, we want to assess the probability that a particular event occurs given partial information. For example, suppose we know the percentage of incoming email messages that are spam. We can determine the likelihood that an incoming email message is spam using the occurrence of words in the message. To determine this likelihood, we need to know the percentage of incoming messages that are spam, the percentage of spam messages in which each of these words occurs, and the percentage of messages that are not spam in which each of these words occurs. To answer this type of questions, we use the following theorem:

Theorem 4 (Bayes' Theorem) Suppose that E and F are events from a sample space such that $p(E) \neq 0$ and $p(F) \neq 0$. Then,

$$p(F | E) = \frac{p(E | F)p(F)}{p(E | F)p(F) + p(E | \bar{F})p(\bar{F})}$$

Proof:

Recall that

$$p(E | F) = \frac{p(E \cap F)}{p(F)}, \quad p(F | E) = \frac{p(F \cap E)}{p(E)}.$$

Since $p(E \cap F) = p(F \cap E)$,

$$p(E | F)p(F) = p(F | E)p(E)$$

Solving for $p(F | E)$,

$$p(F | E) = \frac{p(E | F)p(F)}{p(E)}$$

Now, note that $E = E \cap S = E \cap (F \cup \bar{F}) = (E \cap F) \cup (E \cap \bar{F})$. Note here as well that $(E \cap F)$ and $(E \cap \bar{F})$ are disjoint. Hence, by definition of conditional probability,

$$p(E) = p(E \cap F) + p(E \cap \bar{F}) = p(E | F)p(F) + p(E | \bar{F})p(\bar{F})$$

(This identity is called the law of total probability.)

Therefore,

$$p(F | E) = \frac{p(E | F)p(F)}{p(E | F)p(F) + p(E | \bar{F})p(\bar{F})}$$

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Example 10 Suppose that there are two boxes. The first box contains two green balls and seven red balls. The second contains four green balls and three red balls. Suppose a box is chosen at random, and then a ball is selected from the chosen box at random as well. If a red ball was chosen, what is the probability that the selected ball came from the first box?

Solution: Let E be the event that a red ball was chosen and F be the event that the first box was chosen. Then, $p(F) = \frac{1}{2} = p(\bar{F})$. Also, $p(E | F) = \frac{7}{9}$ and $p(E | \bar{F}) = \frac{3}{7}$. By Bayes' Theorem, the probability that the first box was chosen given that a red ball was chosen is:

$$p(F | E) = \frac{\left(\frac{7}{9}\right)\left(\frac{1}{2}\right)}{\left(\frac{7}{9}\right)\left(\frac{1}{2}\right) + \left(\frac{3}{7}\right)\left(\frac{1}{2}\right)} = \frac{\frac{7}{18}}{\frac{38}{63}} = \frac{49}{76} \approx 0.645$$

Example 11 Most medical tests occasionally produce incorrect results, called false positives and false negatives. A false positive indicates that the test identified that a patient has the disease when they do not actually have it. A false negative result indicates that the test identified that a patient does not have the disease when the patient does have it.

Suppose that one person in 100,000 has a particular disease. There is a test for the disease that gives a positive result 99% of the time when given to someone with the disease. When given to someone without the disease, 99.5% of the time it gives a negative result. Compute for:

1. the probability that a person who tests positive has the disease.
2. the probability that a person who tests negative does not have the disease.

Solution: Let D be the event that a randomly selected person has the disease, and E be the event that a person selected at random tests positive.

1. Need to compute $p(D | E)$ from $p(D)$, $p(E | D)$, $p(E | \bar{D})$ and $p(\bar{D})$.

Since one person in 100,000 is said to have the disease, then $p(D) = \frac{1}{100000} = 0.00001$. By complement rule, $p(\bar{D}) = \frac{99999}{100000} = 0.99999$. From the given, $p(E | D) = 0.99$, hence $p(\bar{E} | D) = 0.01$. Also, $p(\bar{E} | \bar{D}) = 0.995$ implies $p(E | \bar{D}) = 0.005$. Then,

$$\begin{aligned} p(D | E) &= \frac{p(E | D)p(D)}{p(E | D)p(D) + p(E | \bar{D})p(\bar{D})} \\ &= \frac{(0.99)(0.00001)}{(0.99)(0.00001) + (0.005)(0.99999)} \\ &\approx 0.002 \end{aligned}$$

So the chance of the person having a disease given a positive result is very low.

2. Now if the test results are negative,

$$\begin{aligned} p(\bar{D} | \bar{E}) &= \frac{p(\bar{E} | \bar{D})p(\bar{D})}{p(\bar{E} | \bar{D})p(\bar{D}) + p(\bar{E} | D)p(D)} \\ &= \frac{(0.995)(0.99999)}{(0.995)(0.99999) + (0.01)(0.00001)} \\ &\approx 0.9999999 \end{aligned}$$

It is extremely unlikely that a person has the disease when they test negative.

We extend Bayes theorem to any collection of mutually exclusive events that cover the entire sample space S , as follows:

Theorem 5 (Generalized Bayes' Theorem) Suppose that E is an event from a sample space S and $F_1, F_2, \dots, F_n$ are mutually exclusive events such that $\bigcup_{i=1}^n F_i = S$. Assuming that $p(E) \neq 0$ and $p(F_j) \neq 0$ for $j = 1, 2, \dots, n$, then

$$p(F_j | E) = \frac{p(E | F_j)p(F_j)}{\sum_{i=1}^n p(E | F_i)p(F_i)}$$

5 Random Variables and Expected Value

In many problems, we consider a numerical value associated with the outcome of an experiment. For example, the number of ones in a randomly generated string of 10 bits; or the number of times tails come up when a coin is flipped 20 times. Thus, the concept of a random variable is introduced.

Definition 9 A **random variable** is a function from the sample space of an experiment to the set of real numbers. It assigns a real number to each possible outcome. If the range is a subset of the set of integers, then this random variable is said to be a **discrete random variable**.

All throughout our discussions on discrete probability, we are only concerned with discrete random variables.

Example 12 Suppose that a fair coin is flipped three times. The sample space S is the set of possible outcomes $\{HHH, HHT, HTH, THH, TTH, THT, HTT, TTT\}$. Let X be the random variable that assigns to an outcome the number of heads in that outcome. So:

- $X(HHH) = 3$;
- $X(HHT) = X(HTH) = X(THH) = 2$;
- $X(TTH) = X(THT) = X(HTT) = 1$; and,
- $X(TTT) = 0$

Definition 10 The **distribution** of a random variable X on a sample space is the set of pairs $(r, p(X = r))$ for all $r \in X(S)$, where $p(X = r)$ is the probability that X takes the value of r .

Example 13 Each of the eight possible outcomes in Example 12 is equally likely, so each has probability $1/8$. So the distribution of X is determined by the probabilities : $p(X = 3) = \frac{1}{8}$, $p(X = 2) = \frac{3}{8}$, $p(X = 1) = \frac{3}{8}$, and $p(X = 0) = \frac{1}{8}$. Thus, the distribution of X on S is the set of pairs $\{(3, \frac{1}{8}), (2, \frac{3}{8}), (1, \frac{3}{8}), (0, \frac{1}{8})\}$.

Definition 11 The *expected value* (also *expectation* or *mean*) of the random variable X on the sample space S is equal to

$$E(X) = \sum_{s \in S} p(s)X(s)$$

Example 14 Let X be the number that comes up when a fair dice is rolled. What is the expected value of X ?

Solution: The random variable X takes the values 1, 2, 3, 4, 5, and 6, each with probability $1/6$. It follows that

$$E(X) = \frac{1}{6} \cdot 1 + \frac{1}{6} \cdot 2 + \frac{1}{6} \cdot 3 + \frac{1}{6} \cdot 4 + \frac{1}{6} \cdot 5 + \frac{1}{6} \cdot 6 = \frac{21}{6} = \frac{7}{2}$$

Example 15 For the experiment described in Example 12, what is the expected value of X on S ?

Solution: Since each outcome has probability $\frac{1}{8}$, we have

$$\begin{aligned} E(X) &= \frac{1}{8} \{X(HHH) + X(HHT) + X(HTH) + X(THH) + X(TTH) \\ &\quad + X(THT) + X(HTT) + X(TTT)\} \\ &= \frac{1}{8} (3 + 2 + 2 + 2 + 1 + 1 + 1 + 0) \\ &= \frac{12}{8} = \frac{3}{2} \end{aligned}$$

Thus, the expected number of heads that come up when a fair coin is flipped three times is $3/2$.

When an experiment has a large number of outcomes, we can find the expected value of a random variable by grouping together all outcomes assigned the same value by the random variable.

Theorem 6 If X is a random variable and $p(X = r)$ is the probability that $X = r$, so that $p(X = r) = \sum_{s \in S, X(s)=r} p(s)$, then

$$E(X) = \sum_{r \in X(S)} r \cdot p(X = r)$$

Example 16 What is the expected value of the sum of the numbers that appear when a pair of fair dice is rolled?

Solution: Let X be the random variable equal to the sum of the numbers that appear when a pair of dice is rolled. Note that the range of X is $\text{ran}(X) = \{2, 3, 4, \dots, 12\}$. For each $r \in \text{ran}(X)$, $p(X = r)$ is as follows:

- $p(X = 2) = p(X = 12) = \frac{1}{36}$,
- $p(X = 3) = p(X = 11) = \frac{2}{36}$,

- $p(X = 4) = p(X = 10) = \frac{3}{36}$,
- $p(X = 5) = p(X = 9) = \frac{4}{36}$,
- $p(X = 6) = p(X = 8) = \frac{5}{36}$,
- $p(X = 7) = \frac{6}{36}$

Computing for $E(X)$, we have

$$\begin{aligned} E(X) &= 2 \cdot \frac{1}{36} + 3 \cdot \frac{2}{36} + 4 \cdot \frac{3}{36} + 5 \cdot \frac{4}{36} + 6 \cdot \frac{5}{36} + 7 \cdot \frac{6}{36} \\ &\quad + 8 \cdot \frac{5}{36} + 9 \cdot \frac{4}{36} + 10 \cdot \frac{3}{36} + 11 \cdot \frac{2}{36} + 12 \cdot \frac{1}{36} \\ &= 7 \end{aligned}$$

Example 17 Suppose that 500,000 people pay P5 each to play a lottery game with the following prizes: a grand prize of P1,000,000, 10 second prizes of P1,000 each, 1,000 third prizes of P500 each, and 10,000 fourth prizes of P10 each. What is the expected value of a ticket?

Solution: Let X be the random variable equal to the net gain for an individual ticket.

$X(1) = P999,995$ (the net gain for the grand prize ticket, which is one million pesos minus the P5 cost of the winning ticket),

$X(2) = X(3) = \dots = X(11) = P995$ (the net gain for each of the 10 second prize tickets),

$X(12) = X(13) = \dots = X(1011) = P495$ (the net gain for each of the 1,000 third prize tickets), and

$X(1012) = X(1013) = \dots = X(11011) = P5$ (the net gain for each of the 10,000 fourth prize tickets).

Since the remaining 488,989 tickets just lose P5, we have $X(11012) = X(11013) = \dots = X(500000) = -P5$,

Each of the 500,000 lottery tickets has the same chance as any other of containing a winning lottery number, and so $p(k) = \frac{1}{500,000}$ for all $k = 1, 2, 3, \dots, 500000$. Consequently, $p(X = 999995) = \frac{1}{500,000}$, $p(X = 995) = \frac{10}{500,000}$, $p(X = 495) = \frac{1000}{500,000}$, $p(X = 5) = \frac{10000}{500,000}$ and $p(X = -5) = \frac{488989}{500,000}$

The expected value of a ticket is therefore

$$\begin{aligned} E(X) &= \sum_{r \in \text{ran}(X)} r \cdot p(X = r) \\ &= \frac{1}{500000} (999995 \cdot 1 + 995 \cdot 10 + 495 \cdot 1000 + 5 \cdot 10000 + (-5) \cdot 488989) \\ &= -1.78 \end{aligned}$$

In other words, a person who continues to play this lottery for a very long time will probably win some money occasionally but on average will lose 1.78 per ticket.

The next theorem states that expected values are linear, that is, the expected value of the sum of random variables is the sum of their expected values.

Theorem 7 (Linearity of Expectations) *If X_i , $i = 1, 2, \dots, n$ with n a positive integer, are random variables on S , and if a and b are real numbers, then*

$$(i) \ E(X_1 + X_2 + \dots + X_n) = E(X_1) + E(X_2) + \dots + E(X_n)$$

$$(ii) \ E(aX + b) = aE(X) + b.$$

Proof Idea:

(i) For $n = 2$

$$\begin{aligned} E(X_1 + X_2) &= \sum_{s \in S} p(s)(X_1(s) + X_2(s)) \\ &= \sum_{s \in S} p(s)X_1(s) + \sum_{s \in S} p(s)X_2(s) \\ &= E(X_1) + E(X_2) \end{aligned}$$

For $n \geq 2$, prove by mathematical induction.

(ii)

$$\begin{aligned} E(aX + b) &= \sum_{s \in S} p(s)(aX(s) + b) \\ &= a \sum_{s \in S} p(s)X(s) + b \sum_{s \in S} p(s) \\ &= aE(X) + b \quad \left(\text{since } \sum_{s \in S} p(s) = 1 \right) \end{aligned}$$

Example 18 *Find the expected value of the sum of the numbers that appear when a pair of fair dice is rolled.*

Solution: Let X_1 and X_2 be the random variable with $X_1((i, j)) = i$ and $X_2((i, j)) = j$, i.e. X_1 is the number appearing on the first dice, and X_2 is the number appearing on the second dice. From Example 14, $E(X_1) = E(X_2) = \frac{7}{2}$. The sum of the two numbers that appear when two dice are rolled is the sum $X_1 + X_2$. Hence $E(X_1 + X_2) = E(X_1) + E(X_2) = \frac{7}{2} + \frac{7}{2} = 7$. Note that this agrees with the result in Example 16.

Example 19 Hatcheck Problem. *A new employee checks the hats of n people at a restaurant, forgetting to put claim check numbers on the hats. When customers return for their hats, the checker gives them back hats chosen at random from the remaining hats. What is the expected number of hats that are returned correctly?*

Solution: Let X be the random variable that equals the number of people who receive the correct hat from the checker. Let X_i be the random variable with $X_i = 1$ if the i th person receives the correct hat and $X_i = 0$ otherwise. It follows that

$$X = X_1 + X_2 + \dots + X_n$$

Because it is equally likely that the checker returns any of the hats to each person, it follows that the probability that the i th person receives the correct hat is $1/n$. By Theorem 6,

$$E(X_i) = 1 \cdot p(X_i = 1) + 0 \cdot p(X_i = 0) = 1 \cdot \frac{1}{n} + 0 = \frac{1}{n}$$

By linearity of expectations,

$$E(X) = E(X_1) + E(X_2) + \cdots + E(X_n) = n \cdot \frac{1}{n} = 1$$

Hence, the average number of people who receive the correct hat is 1. Note that this result is independent of the number of people who have checked their hats.

We will now define what it means for two random variables to be independent.

Definition 12 *The random variables X and Y on a sample space S are **independent** if*

$$p(X = r_1 \text{ and } Y = r_2) = p(X = r_1) \cdot p(Y = r_2)$$

for all real number r_1 and r_2 .

Example 20 *Consider the example of the two dice sum in Example 18, with the random variable X_1 and X_2 . Are X_1 and X_2 independent?*

Proof: Let $S = \{1, 2, 3, 4, 5, 6\}$ and let $i \in S$ and $j \in S$. Since the 36 possible outcomes of rolling two dice are equally likely,

$$p(X = i \wedge Y = j) = \frac{1}{36}$$

With $p(X = i) = \frac{1}{6}$ and $p(Y = j) = \frac{1}{6}$, then

$$p(X = i) \cdot p(Y = j) = \frac{1}{6} \cdot \frac{1}{6} = \frac{1}{36}$$

Therefore, X_1 and X_2 are independent. ■

Example 21 *From the previous example, show that X_1 and $X = X_1 + X_2$ are not independent.*

Proof: Note that $p(X_1 = 1 \wedge X = 12) = 0$ since $X_1 = 1$ implies that the sum of the numbers appearing on the two dice cannot be equal to 12. Now, $p(X_1 = 1) = \frac{1}{6}$, and $p(X = 12) = \frac{1}{36}$, implying that their product is non-zero. Hence, X_1 and X are not independent. ■

The next theorem states that the expected value of the product of two independent random variables is the product of their expected values.

Theorem 8 *If X and Y are independent random variables, then $E(XY) = E(X)E(Y)$.*

The event $XY = r$ is the disjoint union of the events $X = r_1$ and $Y = r_2$ over all $r_1 \in X(S)$ and $r_2 \in Y(S)$ with $r = r_1 r_2$. Note that if X and Y are not independent, then it cannot be concluded that $E(XY) = E(X)E(Y)$ as illustrated in the next example.

Example 22 Suppose a fair coin is flipped twice. Let X and Y be random variables that count the number of heads and the number of tails respectively. Since $p(X = 2) = 1/4$, $p(X = 1) = 2/4$, and $p(X = 0) = 1/4$, then

$$E(X) = 2 \cdot \frac{1}{4} + 1 \cdot \frac{2}{4} + 0 \cdot \frac{1}{4} = 1$$

Similarly, $E(Y) = 1$. Note that $XY = 0$ when either two heads and no tails or no heads and two tails come up; and $XY = 1$ when one head and one tail come up. Hence,

$$E(XY) = 0 \cdot \frac{2}{4} + 1 \cdot \frac{2}{4} = \frac{1}{2}$$

So $E(XY) \neq E(X)E(Y)$. Note that X and Y are not independent random variables.

6 Variance

The expected value of a random variable tells us the average value, but nothing about how widely its values are distributed. The variance of a random variable measures how wide the distribution of X is about its expected value.

Definition 13 The **deviation** of X at $s \in S$ is $X(s) - E(X)$, the difference between the value of X at s and the mean of X .

Definition 14 Let X be a random variable on sample space S . The **variance** of X , denoted by $V(X)$, is

$$V(X) = \sum_{s \in S} (X(s) - E(X))^2 p(s)$$

That is, $V(X)$ is the weighted average of the square of the deviation of X . The **standard deviation** of X , denoted as $\sigma(X)$, is defined as $\sqrt{V(X)}$.

Example 23 What is the variance of a random variable X , where X is the number that comes up when a fair dice is rolled?

Solution: We have $V(X) = E(X^2) - E(X)^2$. From Example 14, $E(X) = 7/2$. Also, $p(s) = 1/6$ for every $s \in \{1, 2, \dots, 6\}$. Hence,

$$\begin{aligned} V(X) &= \left(\frac{1}{6}\right) \left\{ \left(1 - \frac{7}{2}\right)^2 + \left(2 - \frac{7}{2}\right)^2 + \left(3 - \frac{7}{2}\right)^2 + \left(4 - \frac{7}{2}\right)^2 + \left(5 - \frac{7}{2}\right)^2 + \left(6 - \frac{7}{2}\right)^2 \right\} \\ &= \frac{35}{12} \end{aligned}$$

Theorem 9 If X is random variable on a sample space S , then $V(X) = E(X^2) - E(X)^2$.

Corollary 1 If X is a random variable on a sample space S and $E(X) = \mu$, then $V(X) = E((X - \mu)^2)$.

Example 24 Verify Theorem 9 for Example 23.

Solution: We have $E(X) = 7/2$. Also, $p(s) = 1/6$ for every $s \in \{1, 2, \dots, 6\}$. Hence,

$$E(X^2) = \frac{1}{6}(1^2 + 2^2 + 3^2 + 4^2 + 5^2 + 6^2) = \frac{91}{6}$$

and

$$V(X) = \frac{91}{6} - \left(\frac{7}{2}\right)^2 = \frac{35}{12}$$

Example 25 What is the variance of the random variable $X(i, j) = 2i$, where i is the number appearing on the first dice and j is the number appearing on the second dice, when two fair dice are rolled?

Solution: First, find the $E(X)$ and $E(X^2)$. Note that $p(X = k)$ is $1/6$ for $k = 2, 4, 6, 8, 10, 12$ and is 0 otherwise. So,

$$E(X) = \frac{1}{6}(2 + 4 + 6 + 8 + 10 + 12) = 7$$

and

$$E(X^2) = \frac{1}{6}(2^2 + 4^2 + 6^2 + 8^2 + 10^2 + 12^2) = \frac{182}{3}$$

Then by Theorem 9,

$$V(X) = \frac{182}{3} - 49 = \frac{35}{3}$$

Theorem 10 (Bienaymé's Formula) If X and Y are two independent random variables on sample space S , then $V(X + Y) = V(X) + V(Y)$. Furthermore if $X_i, i = 1, 2, \dots, n$, with n a positive integer, are pairwise independent variables on S , then, $V(X_1 + X_2 + \dots + X_n) = V(X_1) + V(X_2) + \dots + V(X_n)$.

Example 26 Find the variance and standard deviation of the random variable X whose value when two fair dice are rolled is $X((i, j)) = i + j$, where i is the number on the first dice, and j is the number on the second dice.

Solution: Let X_1 and X_2 be random variables such that $X_1((i, j)) = i$ and $X_2((i, j)) = j$. Then $X = X_1 + X_2$, and X_1 and X_2 are independent as in Example 20. It follows that $V(X) = V(X_1) + V(X_2)$. Note that

$$E(X_1) = \frac{1}{6}(1 + 2 + 3 + 4 + 5 + 6) = \frac{7}{2}$$

and

$$E(X_1^2) = \frac{1}{6}(1^2 + 2^2 + 3^2 + 4^2 + 5^2 + 6^2) = \frac{91}{6}$$

Hence,

$$V(X_1) = \frac{91}{6} - \frac{49}{4} = \frac{35}{12}$$

Similarly, we get the same values for X_2 , that is, $V(X_1) = V(X_2) = 35/12$. Hence,

$$V(X) = \frac{35}{12} + \frac{35}{12} = \frac{35}{6}$$

and the standard deviation is $\sigma(X) = \sqrt{35/6}$.

Now, how about $V(X - Y)$? How about $V(aX + b)$?

Show that if X is a random variable on a sample space S , then $V(aX + b) = a^2V(X)$ whenever a and b are real numbers. Also, show that if X and Y are two independent random variables on S , then $V(X - Y) = V(X) + V(Y)$.

7 Bernoulli Trials

Definition 15 Suppose that an experiment can have only two possible outcomes (e.g. flipping a coin yields only either a head or tail).

- Each performance of the experiment is called a **Bernoulli Trial**.
- One outcome is called a **success** and the other a **failure**.
- If p is the probability of success and q the probability of failure, then $p + q = 1$.

Bernoulli trials are mutually independent if the conditional probability of success on any given trial is p , given any information whatsoever about the outcomes of the other trials. Now, consider the following example:

Example 27 A coin is biased so that the probability of heads is 0.6. Suppose the coin is tossed ten times. It is reasonable to assume that results of the tosses are mutually independent.

1. What is the probability of obtaining exactly eight heads?
2. What is the probability of obtaining at least eight heads?

Solution: For each $i = 1, \dots, 10$, let H_i be the event that a head is obtained at the i th toss, while T_i be the event that a tail is obtained on the i th toss.

Suppose that eight heads occur on the first eight tosses and the two remaining tosses are tails. This event is

$$H_1 \cap H_2 \cap H_3 \cap H_4 \cap H_5 \cap H_6 \cap H_7 \cap H_8 \cap T_9 \cap T_{10}$$

We can denote this as $HHHHHHHHTT$. By definition of mutually independent events, $p(HHHHHHHHTT) = (0.6)^8(0.4)^2$. Note that if the eight heads occur on any of the ten tosses, the same probability is obtained.

Now, there would be $\binom{10}{8}$ ways in which eight heads occur in these ten tosses (permutation of indistinguishable objects). Each way of obtaining eight heads is mutually exclusive. Hence,

$$p(\text{eight heads}) = \binom{10}{8}(0.6)^8(0.4)^2$$

With similar reasoning,

$$p(\text{nine heads}) = \binom{10}{9}(0.6)^9(0.4)$$

and

$$p(\text{ten heads}) = \binom{10}{10}(0.6)^{10}$$

Hence, the probability of obtaining at least eight heads is:

$$\begin{aligned} p(\text{at least eight heads}) &= p(\text{eight heads}) + p(\text{nine heads}) + p(\text{ten heads}) \\ &= \binom{10}{8}(0.6)^8(0.4)^2 + \binom{10}{9}(0.6)^9(0.4) + \binom{10}{10}(0.6)^{10} \\ &\approx 0.167 \end{aligned}$$

Following the same reasoning as was used in this, the probability of exactly k successes in n mutually independent Bernoulli trials is given in the next theorem.

Theorem 11 *The probability of exactly k successes in n mutually independent Bernoulli trials, with probability of success p is*

$$\binom{n}{k} p^k (1-p)^{n-k}.$$

Note: The sum of the probabilities that there are k successes when n independent Bernoulli trials are carried out, for $k = 0, 1, 2, \dots, n$, equals

$$\sum_{k=0}^n \binom{n}{k} p^k (1-p)^{n-k} = (p + (1-p))^n = 1, \text{ as should be the case.}$$

Theorem 12 *The expected number of successes when n mutually independent Bernoulli trials are performed, where p is the probability of success on each trial, is np .*

Proof: Let X be the random variable equal to the number of successes in n trials. We have

$$\begin{aligned} E(X) &= \sum_{k=1}^n k \cdot p(X = k) && \text{by Theorem 6} \\ &= \sum_{k=1}^n k \binom{n}{k} p^k (1-p)^{n-k} && \text{by Theorem 11} \\ &= \sum_{k=1}^n n \binom{n-1}{k-1} p^k (1-p)^{n-k} && \text{since } k \binom{n}{k} = n \binom{n-1}{k-1} \\ &= np \sum_{k=1}^n \binom{n-1}{k-1} p^{k-1} (1-p)^{n-k} \\ &= np \sum_{j=0}^{n-1} \binom{n-1}{j} p^j (1-p)^{n-1-j} \\ &= np(p + (1-p))^{n-1} && \text{by the binomial theorem} \\ &= np \end{aligned}$$

■

Example 28 *Give an alternative proof for Theorem 12 using linearity of expectations.*

Proof: Let X_i be the random variable with $X_i((t_1, \dots, t_n)) = 1$ if t_i is a success and $X_i((t_1, \dots, t_n)) = 0$ if t_i is a failure. The expected value of X_i is $E(X_i) = 1 \cdot p + 0 \cdot (1-p) = p$, for all $i = 1, \dots, n$. Let $X = X_1 + \dots + X_n$, so that X counts the number of successes when these n Bernoulli trials are performed. Hence, by Theorem 7, $E(X) = E(X_1) + \dots + E(X_n) = np$.

■

Example 29 What is the variance of the random variable X with $X(t) = 1$ if a Bernoulli trial is a success and $X(t) = 0$ if it is a failure, where p is the probability of success and q is the probability of failure?

Solution: The expected value of X is $E(X) = 0 \cdot (1 - p) + 1 \cdot p = p$. Also, because X takes only the values 0 and 1, then $[X(t)]^2 = X(t)$. Hence,

$$V(X) = E(X^2) - E(X)^2 = p - p^2 = p(1 - p) = pq$$

Example 30 What is the variance of the number of successes when n mutually independent Bernoulli trials are performed, where on each trial, p is the probability of success and q is the probability of failure?

Solution: Let X_i be the random variable with $X_i((t_1, t_2, \dots, t_n)) = 1$ if trial t_i is a success and $X_i((t_1, t_2, \dots, t_n)) = 0$ if trial t_i is a failure. Let $X = X_1 + X_2 + \dots + X_n$. From the previous example, $V(X_i) = pq$, for all $i = 1, 2, \dots, n$. Since each X_i is independent of each other, $V(X) = V(X_1) + V(X_2) + \dots + V(X_n) = np(1 - p) = npq$.

8 Discrete Distributions

Assume X is a discrete random variable. We would like to specify probabilities of events $\{X = r\}$. Recall that if we can specify the probabilities involving X , we can say that we have specified the probability distribution of X . Also, for a countable set of values $r_1, r_2, \dots, r_n$, we have $p(X = r_i) > 0, i = 1, 2, \dots, n$ and $\sum_i p(X = r_i) = 1$.

We can then define the **probability mass function (pmf)** f of X by $f(r) = p(X = r)$, sometimes written as f_X .

Definition 16 A random variable X has a **uniform distribution** with $n = |\text{ran}(X)|$, if $p(X = k) = \frac{1}{n}$, for all $1 \leq k \leq n$.

Example 31 Let X be the number that comes up when a fair dice is rolled (as in Example 14). X has range $\{1, 2, 3, 4, 5, 6\}$. The pmf is

r	1	2	3	4	5	6
$f(r)$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$

Definition 17 A random variable X for a Bernoulli trial with probability of success p , has a **Bernoulli distribution** if X takes just two possible values, 0 and 1, with $p(X = 1) = p$ and $p(X = 0) = 1 - p$. Its parameter is p , the probability of success.

Theorem 13 If the random variable X has a Bernoulli distribution with parameter p , then

$$E(X) = p \text{ and } V(X) = p(1 - p)$$

Proof: The pmf is

r	0	1
$f(r)$	$1 - p$	p

Hence,

$$\begin{aligned}
 E(X) &= \sum_r r \cdot p(X = r) \\
 &= 0 \cdot p(X = 0) + 1 \cdot p(X = 1) \\
 &= 0 \cdot (1 - p) + 1 \cdot p \\
 &= p \\
 V(X) &= \sum_r (r - p)^2 \cdot p(X = r) \\
 &= (0 - p)^2 \cdot p(X = 0) + (1 - p)^2 \cdot p(X = 1) \\
 &= (0 - p)^2(1 - p) + (1 - p)^2 p \\
 &= p(1 - p)[p + (1 - p)] = p(1 - p)
 \end{aligned}$$

Note that this agrees with the result in Example 29

Definition 18 A random variable X described as the number of successes in a sequence of mutually independent Bernoulli trials has a **binomial distribution**. Its parameters are n , the number of trials, and p , the probability of success. We write it as $X \sim \text{Bin}(n, p)$.

Theorem 14 If the random variable X has a Binomial distribution, that is, $X \sim \text{Bin}(n, p)$, then its pmf is

$$f(r) = p(X = r) = \binom{n}{r} p^r (1 - p)^{n-r} \text{ with } r = 0, 1, \dots, n.$$

Also,

$$E(X) = np \text{ and } V(X) = np(1 - p)$$

Proof: Refer to Theorem 11, Theorem 12 and Example 30.

Definition 19 A random variable X described as the number of independent Bernoulli trials needed to get the first success has a **geometric distribution** with parameter p , the probability of success, in each trial. We write it as $X \sim G(p)$.

Example 32 Suppose that the probability that a coin comes up tails is p . This coin is flipped repeatedly until it comes up tails. What is the expected number of flips until this coin comes up tails?

Solution: The sample space consists of sequences that begin with any number of heads, H , then followed by a tail, T . Hence, the sample space is $\{T, HT, HHT, HHHT, \dots\}$, which is infinite. The probability of obtaining a head is $(1 - p)$. So $p(T) = p$, $p(HT) = (1 - p)p$, $p(HHT) = (1 - p)^2 p$, and so on. In general, the probability of flipping the coin k times before a tail comes out is $(1 - p)^{k-1} p$.

Now, let the random variable X be the number of coin flips in an element in the sample space, i.e. $X(T) = 1$, $X(HT) = 2$, and so on. Note as well that $p(X = k)$ is $(1 - p)^{k-1}p$. By Theorem 6,

$$\begin{aligned} E(X) &= \sum_{k=1}^{\infty} k \cdot p(X = k) = \sum_{k=1}^{\infty} k \cdot (1 - p)^{k-1}p \\ &= p \cdot \sum_{k=1}^{\infty} k \cdot (1 - p)^{k-1} = p \cdot \left(\frac{1}{(1 - (1 - p))^2} \right) \\ &= p \cdot \left(\frac{1}{p^2} \right) = \frac{1}{p} \end{aligned} \quad (*)$$

* Note, the identity (see generating functions)

$$\sum_{k=1}^{\infty} k \cdot x^{k-1} = \frac{1}{(1 - x)^2}$$

Theorem 15 *If the random variable X has a geometric distribution with parameter p , then its pmf is*

$$f(r) = p(X = r) = (1 - p)^{r-1}p \quad \text{for all integer } r \geq 1.$$

Also,

$$E(X) = \frac{1}{p} \text{ and } V(X) = \frac{1 - p}{p^2}.$$

Proof: Refer to Example 32 for pmf and $E(X)$. We do a similar proof for $V(X)$.

$$\begin{aligned} E(X^2) &= \sum_{k=1}^{\infty} k^2 \cdot p(X = k) = \sum_{k=1}^{\infty} k^2 \cdot (1 - p)^{k-1}p \\ &= p \cdot \sum_{k=1}^{\infty} k^2 \cdot (1 - p)^{k-1} = p \cdot \left(\frac{1 + (1 - p)}{(1 - (1 - p))^3} \right) \\ &= p \cdot \left(\frac{2 - p}{p^3} \right) = \frac{2 - p}{p^2} \end{aligned} \quad (**)$$

** Note, the identity (see generating functions)

$$\sum_{k=1}^{\infty} k^2 \cdot x^{k-1} = \frac{1 + x}{(1 - x)^3}$$

Hence,

$$V(X) = E(X^2) - (E(X))^2 = \frac{2 - p}{p^2} - \left(\frac{1}{p} \right)^2 = \frac{1 - p}{p^2}$$

■

Definition 20 *A random variable X for which $p(X = x) = \frac{\lambda^x}{x!} e^{-\lambda}$, $x = 0, 1, 2, \dots$ for fixed $\lambda > 0$ has a **Poisson distribution**, written as $X \sim \text{Poi}(\lambda)$.*

Theorem 16 *If the random variable X has a Poisson distribution with parameter λ , then*

$$E(X) = \lambda \text{ and } V(X) = \lambda.$$

Proof:

$$\begin{aligned} E(X) &= \sum_{k=0}^{\infty} k \cdot p(X = k) \\ &= \sum_{k=0}^{\infty} k \cdot \frac{\lambda^k}{k!} e^{-\lambda} = 0 + \sum_{k=1}^{\infty} k \cdot \frac{\lambda^k}{k!} e^{-\lambda} \\ &= \lambda \cdot e^{-\lambda} \cdot \sum_{k=1}^{\infty} \frac{\lambda^{k-1}}{(k-1)!} = \lambda \cdot e^{-\lambda} \cdot e^{\lambda} \quad (***) \\ &= \lambda \end{aligned}$$

*** Note, the identity (see generating functions)

$$\sum_{k=1}^{\infty} \frac{x^{k-1}}{(k-1)!} = \sum_{k=0}^{\infty} \frac{x^k}{k!} = e^x$$

Similarly,

$$\begin{aligned} E(X^2) - E(X) &= \sum_{k=0}^{\infty} (k^2 - k) \cdot p(X = k) \\ &= \sum_{k=0}^{\infty} (k^2 - k) \cdot \frac{\lambda^k}{k!} e^{-\lambda} = 0 + \sum_{k=2}^{\infty} k(k-1) \cdot \frac{\lambda^k}{k!} e^{-\lambda} \\ &= \lambda^2 \cdot e^{-\lambda} \cdot \sum_{k=2}^{\infty} \frac{\lambda^{k-2}}{(k-2)!} = \lambda^2 \cdot e^{-\lambda} \cdot e^{\lambda} \quad (****) \\ &= \lambda^2 \end{aligned}$$

**** Note, the identity (see generating functions)

$$\sum_{k=2}^{\infty} \frac{x^{k-2}}{(k-2)!} = \sum_{k=0}^{\infty} \frac{x^k}{k!} = e^x$$

Hence,

$$V(X) = E(X^2) - (E(X))^2 = (\lambda^2 + E(X)) - (E(X))^2 = \lambda^2 + \lambda - \lambda^2 = \lambda$$

Theorem 17 *A Poisson distribution $Poi(\lambda)$ can be seen as a limiting distribution of $Bin(n, p)$ where $\lambda = np$.*

That is, Poisson distribution can be effectively used to approximate Binomial distribution when the number of trials n is large, and the probability of success p is small. Such an approximation is adequate, say, for $n \geq 30$ and $p \leq 0.5$, and it becomes more accurate for larger n .

Proof: Mathematically, it means $Bin(n, p) \approx Poi(\lambda)$ where $\lambda = np, n \rightarrow \infty, p \rightarrow 0$, that is,

$$\lim_{\substack{n \rightarrow \infty \\ p \rightarrow 0 \\ np \rightarrow \lambda}} \binom{n}{x} p^x (1-p)^{n-x} = \frac{\lambda^x}{x!} e^{-\lambda}$$

9 The Chebyshev's Inequality and The Law of Large Numbers

How likely is it that a random variable takes a value far from its expected value? The next theorem, Chebyshev's inequality, helps answer this question by providing an upper bound on the probability that the value of a random variable differs from the expected value of the random variable by more than a specified amount.

Theorem 18 (Chebyshev's Inequality) *Let X be a random variable on sample space S with probability function p . If r is a positive real number, then*

$$p(|X(s) - E(X)| \geq r) \leq \frac{V(X)}{r^2}$$

Proof: Let A be the event that the value of X differ from the mean $E(X)$ by at least r . That is,

$$A = \{s \in S \mid |X(s) - E(X)| \geq r\}.$$

Note,

$$\begin{aligned} V(X) &= \sum_{s \in S} (X(s) - E(X))^2 p(s) \\ &= \sum_{s \in A} (X(s) - E(X))^2 p(s) + \sum_{s \notin A} (X(s) - E(X))^2 p(s) \\ &\geq \sum_{s \in A} (X(s) - E(X))^2 p(s) \\ &\quad \text{since each of the summands in the second sum is nonnegative} \\ &\geq \sum_{s \in A} r^2 p(s) = r^2 \sum_{s \in A} p(s) = r^2 p(A) \quad \text{by definition of } A \end{aligned}$$

Hence, it follows that $p(A) \leq \frac{V(X)}{r^2}$. ■

Example 33 *Suppose that X is the random variable that counts the number of tails when a fair coin is tossed n times. Note that X is the number of successes when n independent Bernoulli trials, each with probability of success $1/2$ are done. Hence, by Theorem 14, $E(X) = n/2$ and $V(X) = n/4$.*

By Chebyshev's inequality with $r = \sqrt{n}$,

$$p(|X(s) - n/2| \geq \sqrt{n}) \leq \frac{n/4}{(\sqrt{n})^2} = \frac{1}{4}$$

This means that the probability that the number of tails that come up on n tosses deviates from the mean by more than $\sqrt{n}$ is no more than $\frac{1}{4}$.

The Chebyshev's inequality is applicable to any random variable, but it often fails to give a practical estimate for the probability that the value of a random variable exceeds its mean by a large amount. This is illustrated in the next example.

Example 34 *Let X be the random variable whose value is the number that comes up when a fair die is rolled. We have $E(X) = 7/2$ (refer to Example 14) and $V(X) = 35/12$ (refer to Example 23). Because the only possible values of X are in $\{1, 2, 3, 4, 5, 6\}$, X cannot take a value more than $5/2$ from its mean, $E(X)$. Hence, $p(|X - 7/2| \geq r) = 0$ if $r > 5/2$. By Chebyshev's Inequality, we have $p(|X - 7/2| \geq r) \leq \frac{35/12}{r^2}$.*

Take $r = 3$, from Chebyshev's Inequality, we have $p(|X - 7/2| \geq 3) \leq \frac{35/12}{9} = 35/108 \approx 0.324$. This is not a good estimate of the real value $p(|X - 7/2| \geq 3) = 0$.

The next theorem describes how when an experiment is repeated, independent to each other, a large number of times, the average of the result obtained should be close to the expected value, and will tend to become closer as more trials are performed.

Theorem 19 (Law of Large Numbers) *Let $X_1, X_2, \dots, X_n$ be a sequence of independent and identically distributed random variables, with finite expected value $\mu = E(X_j)$ and finite variance $\sigma^2 = V(X_j)$. Let $S_n = X_1 + X_2 + \dots + X_n$. Then for any $\epsilon > 0$,*

$$p\left(\left|\frac{S_n}{n} - \mu\right| \geq \epsilon\right) \rightarrow 0$$

as $n \rightarrow \infty$.

Equivalently,

$$p\left(\left|\frac{S_n}{n} - \mu\right| < \epsilon\right) \rightarrow 1$$

as $n \rightarrow \infty$.

Proof: Since $X_1, X_2, \dots, X_n$ are independent and have the same distributions, we have $E\left(\frac{S_n}{n}\right) = \mu$. Also, $V(S_n) = n\sigma^2$ and $V\left(\frac{S_n}{n}\right) = \frac{\sigma^2}{n}$.

By Chebyshev's inequality, for any $\epsilon > 0$,

$$p\left(\left|\frac{S_n}{n} - \mu\right| \geq \epsilon\right) \leq \frac{\sigma^2}{n\epsilon^2}$$

It follows that as $n \rightarrow \infty$,

$$p\left(\left|\frac{S_n}{n} - \mu\right| \geq \epsilon\right) \rightarrow 0.$$

■

Note that $\frac{S_n}{n}$ is an average of the individual outcomes. The law implies that the sample average converges to the expected value as the sample size increases. Basically, starting with a random experiment about which little can be predicted and, by taking averages, we obtain an experiment in which the outcome can be predicted with a high degree of certainty. The Law of Large Numbers, as have been stated, is often called the “Weak Law of Large Numbers”.

Example 35 *Consider the special case of Bernoulli trials with probability p for success. Let $X_j = 1$ if the j th outcome is a success and $X_j = 0$ if it is a failure. Then $S_n = X_1 + X_2 + \dots + X_n$ is the number of successes in n trials and $\mu = E(X_j) = p$. The Law of Large Numbers states that for any $\epsilon > 0$*

$$p\left(\left|\frac{S_n}{n} - p\right| < \epsilon\right) \rightarrow 1$$

as $n \rightarrow \infty$.

The above statement says that, in a large number of repetitions of a Bernoulli experiment, we can expect the proportion of times the event will occur to be near p . This shows that our mathematical model of probability agrees with our frequency interpretation of probability.